

The Impact of Wildfire Evacuation on Rental Prices: Evidence from Canada

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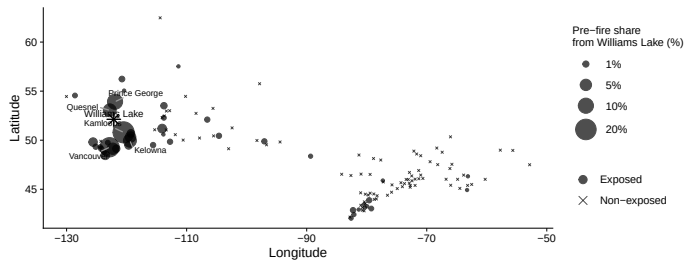
The main finding

Cities more connected to Williams Lake show **higher rents** after the 2017 wildfire.

Specification	$\hat{\pi}$	Implied effect (Kamloops)
Baseline TWFE	0.818*** (0.216)	+19% / +\$160/month
City trends	0.594*** (0.175)	+13%
Province $\times$ Year FE	0.164 (0.260)	indistinguishable from zero

Credible range: $\hat{\pi} \in [0.164, 0.818]$

The question



*July 2017. Williams Lake, BC: 11,000 evacuees, <24 hours' notice.
Where do they go? Do rents rise there?*

Identification strategy

Exposure index (shift-share, continuous DiD):

$$Z_{WL,dt} = \underbrace{s_{WL,d}}_{\text{pre-fire share}} \times \underbrace{1[t \geq 2017]}_{\text{post-fire}}$$

Main regression:

$$\log R_{dt} = \alpha + \pi Z_{WL,dt} + \gamma_d + \lambda_t + \varepsilon_{dt}$$

$s_{WL,d}$ = share of WL out-migrants settling in city d , wave 2016/2017

γ_d = city fixed effects λ_t = year fixed effects

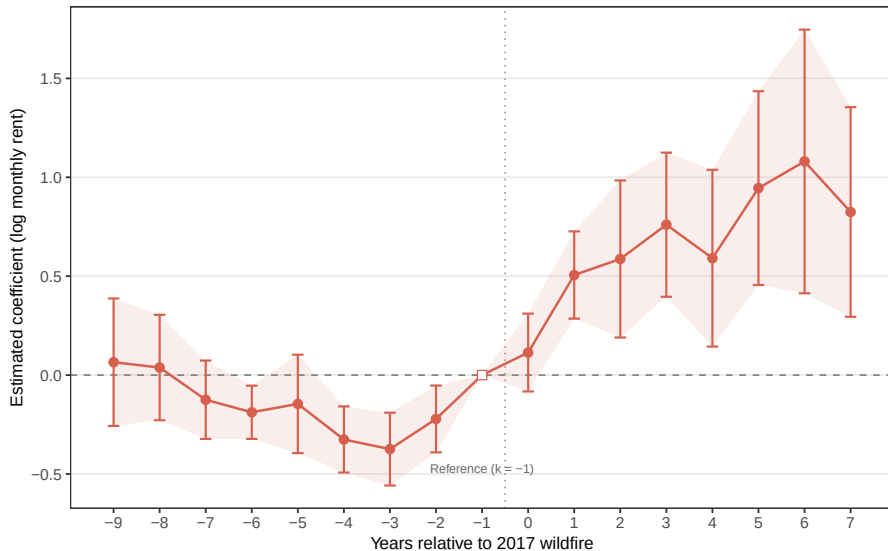
Panel: 131 Canadian cities, 2008–2024. SE clustered by city.

Main results: five specifications

	(1) OLS	(2) +Year FE	(3) +City FE	(4) TWFE	(5) +City trends
Z_{WL}	3.84*** (1.05)	2.49*** (0.86)	3.40*** (0.85)	0.818*** (0.216)	0.594*** (0.175)
N	2,225	2,225	2,225	2,225	2,225
City FE			✓	✓	✓
Year FE		✓		✓	✓
City trends					✓

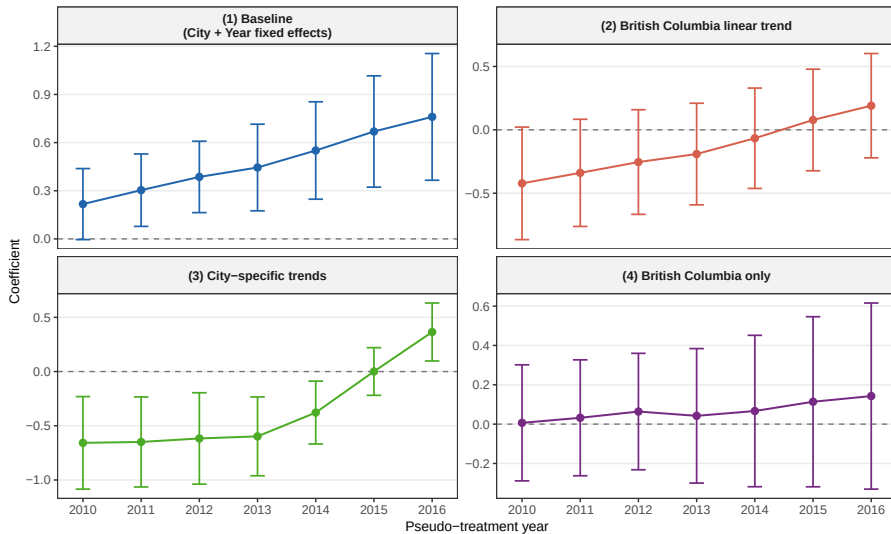
Dep. var.: log monthly rent, two-bedroom apartments. SE clustered by city.

Dynamic effects: event study



Pre-trends near zero. Coefficients grow from $k = 1$ onward. At $k = 6$: $\hat{\beta}_6 \approx 1.22$ log-points ($\approx 29\%$ for

The identification challenge



Problem: placebo treatment years 2010–2016 produce positive coefficients in the baseline.

Province $\times$ Year FE absorbs BC dynamics: $\hat{\alpha} = 0.164$ (s.e. 0.260), not significant

The Sudbury result

Origin	Coefficient	SE
Williams Lake (pre-COVID, ≤ 2019)	0.544***	(0.177)
Greater Sudbury (placebo, Ontario)	0.548***	(0.208)
Quesnel (placebo, near WL)	0.682*	(0.348)

Greater Sudbury: 5,000 km from Williams Lake. No wildfire. Orthogonal migration network.

$\Rightarrow$ The design captures **migration connectivity**, not the specific wildfire event.

What the data say

Test	Estimate
Baseline TWFE	0.818***
Province $\times$ Year FE	0.164 (not sig.)
Credible range	[0.164, 0.818]

Pre-existing migration networks **predict** post-2017 rent trajectories.

Provincial dynamics confound the wildfire-specific reading.

Annual migration flows show **no** post-fire amplification: -95.8^{***} (s.e. 8.9).

Policy and next steps

Policy implication:

Disaster relief stops at the burn perimeter. Receiving cities bear housing costs that the current framework ignores.

To establish a causal estimate, the next design needs:

Limitation	Solution
Single origin (WL)	Multiple wildfires, multiple origins
BC regional confounding	Cross-provincial variation in exposure
Annual migration data	Higher-frequency mobility records

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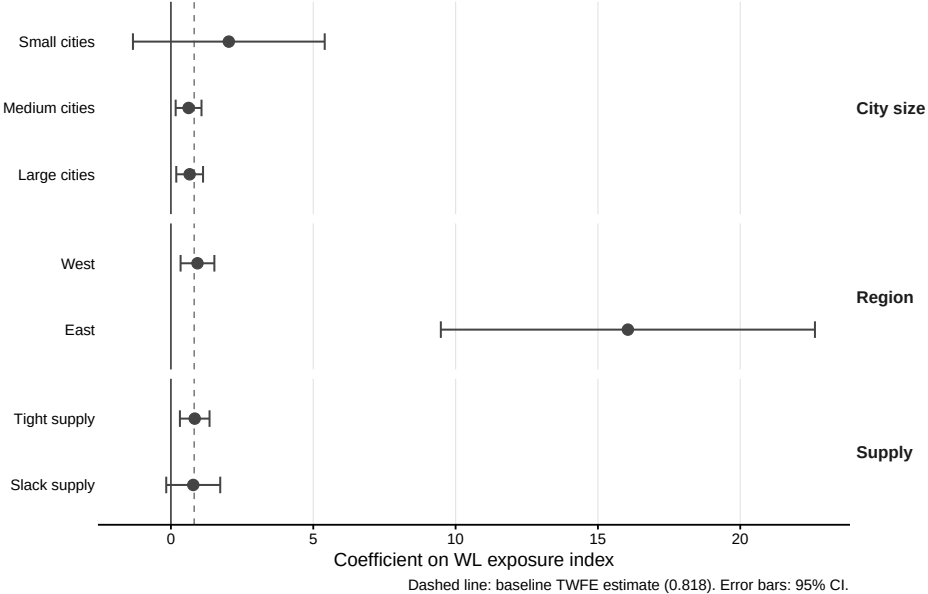
Evidence from Canada — Williams Lake, July 2017

Setting	131 Canadian cities, 2008–2024
Design	Continuous DiD, shift-share exposure
Baseline estimate	$\hat{\pi} = 0.818^{***}$ (+19% for Kamloops)
Credible range	[0.164, 0.818]
Channel test	No migration amplification (-95.8^{***})
Reading	Reduced-form correlation along networks

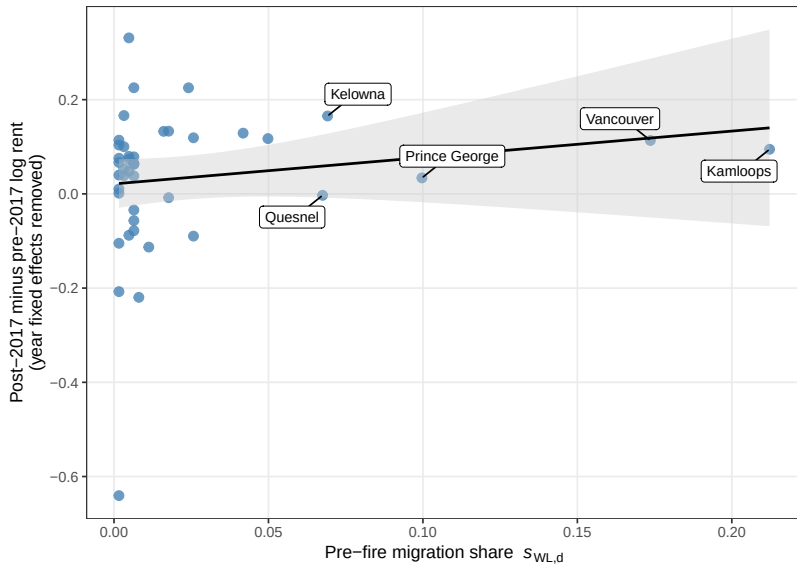
Code & data:

github.com/Robin-Economist/The-Impact-of-Wildfires-on-Rental-Prices-Evidence-from-Canada

Appendix A — Heterogeneity by city size, region, supply



Appendix B — Reduced-form scatter



Slope = 0.818. Each point is an exposed city ($s_{WL,d} > 0$, $N = 43$)

Appendix C — Economic calibration

City	Share $s_{WL,d}$	$\Delta\%$ baseline	$\Delta\$$ baseline
Kamloops	21.2%	19.0%	\$161
Vancouver	17.4%	15.3%	\$129
Prince George	10.0%	8.5%	\$72
Kelowna	6.9%	5.8%	\$49
Quesnel	6.8%	5.7%	\$48

Reference rent: \$848 (national average, 2015). Baseline $\hat{\pi} = 0.818$.

Appendix D — Rambachan-Roth sensitivity

Breakdown value $\bar{M}^* = 0.38$: the post-2017 confidence set ceases to exclude zero once post-treatment trend violations reach 38% of the largest pre-treatment deviation.

